

PARTS LIST

Semiconductors:

IC1	- 7805 voltage regulator
IC2	- ATtiny85 micro controller
D1, D2	- 1N4007 rectifier diode
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R5	- 2.2-kilo-ohm
R2	- 47-kilo-ohm
R3, R4	- 10-kilo-ohm
VR1	- 10-kilo-ohm potmeter

Capacitors:

C1	- 10 μ F, 25V electrolytic
C2	- 10 μ F, 16V electrolytic
C3	- 0.1 μ F ceramic disk

Miscellaneous:

Batt.1	- 9V battery
Batt.2	- 12V, 7Ah-30Ah battery
CON1	- 3-pin connector
CON2	- 4-pin connector
S1	- On/off switch
	- ACS712 module, 30A
	- OLED, 2.4384 cm, 128x64pixel
	- 8-pin IC socket
	- Jumper wires
	- Heat-sink for 7805

(All these are available from kitsnspares.com)

Accuracy of ACS712 may slightly vary from module to module. Multiplication factor is initialised as 740 in the source program, which is achieved from 66mV/A (as per Allegro's data sheet). The value may be varied in the source code (vaw_meter.c), declared as int MF = 740.

Then, re-compile the code using AVR studio and upload vaw_meter.hex file to the MCU.

Construction and testing

After assembling the circuit on the PCB or veroboard, burn vaw_meter.hex to ATtiny85 MCU using any AVR programmer. Insert the programmed ATtiny85 to its eight-pin socket/base. Connect the OLED (four-pin) to the circuit at connector CON2. Power on the board with 9V battery (Batt.1) via switch S1.

Power indicator LED1 will be on. LED2 will start blinking, indicating that program initialisation is complete and the circuit is ready for use. Now, the OLED will start displaying volts, amps and watts after a short delay.

Next, connect 12V battery (Batt.2) at DC input pins. Adjust VR1 to display the same voltage on the OLED. Connect jumper wires J2 and J3 externally to positive DC output and ground, respectively. J1 is optional and should be connected to positive DC input only when 8V DC to 20V DC power supply is used.

Connect 12V bulb (used in automobiles) as load or any load at DC output pins. The OLED displays current and power consumption. The + and - symbols in the OLED display prefixed to AMPS indicate the direction of current with respect to ACS712 module terminals (IP+ and IP-), as shown in the circuit.

In case 8V to 20V is used for measurement, connect J1 to positive DC input to get the power supply to the circuit from 12V battery. Remove 9V battery in that case. **EFY**



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** Can be increased further, if needed.



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